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[]: Question no 1

```
[51]: def calculate_average(num1, num2, num3):
```

```
    return (num1+ num2+ num3) / 3
```

```
avg = calculate_average(10, 50, 30)
```

```
print(f"average: {avg}")
```

average: 30.0

[]: Question no 2

```
[44]: from datetime import datetime
```

```
def greet_with_time(name):
```

```
    current_time = datetime.now().strftime("%H:%M:%S")
```

```
    print(f"hello, {name}! this current time is {current_time}.")
```

```
greet_with_time("sana")
```

hello, sana! this current time is 17:39:19.

Question no 3

```
[53]: def calculate_discount(price, discount=10):
```

```
    discount_amount = price * (discount / 100)
```

```
    return price - discount_amount
```

```
print(f"default (20%): {calculate_discount(100)}")
```

```
print(f"costum (35%): {calculate_discount(100, 25)}")
```

default (20%): 90.0

costum (35%): 75.0

[]: Question no 4

```
[55]: def print_info(name, age, city):
```

```
    print(f"name: {name}, age: {age}, city:{city}")
```

```
print_info(name="saba", age=21, city="lahore")
```

namme: saba, age: 21, city:lahore

[]: Question no 5

```
[47]: def find_max(* args):  
        if not args:  
            return none  
            return max(args)  
  
print(f"max value: {find_max(5, 12, 8, 3)}")
```

max value: None

[]: Question no 06

```
[48]: def factional(n):  
        if n == 0 or n == 1:  
            return 1  
        else:  
            return n * factional(n-1)  
  
print(f"factional of 5: {factional(5)}")
```

factional of 5: 120

```
[49]: def apply_function(func, numbers):  
        return [func(n) for n in numbers]  
  
def square(x):  
    return x * x  
  
nums = [1, 2, 3, 4]  
result = apply_function(square, nums)  
print(f"squared list: {result}")
```

squared list: [1, 4, 9, 16]